Tensors and SE(3)

Topics

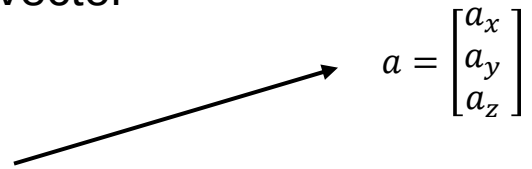
- Tensors Review
- Coordinate Frames in SE(3)
- Position and Orientation in SE(3)
- Homogenous Transforms

Complementary Reading: J.J. Craig, Introduction to Robotics, Chapter 2

Tensor Preliminaries

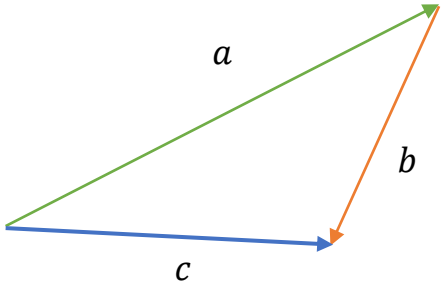
Vector: i.e. always assume to be a column vector

$$a = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$



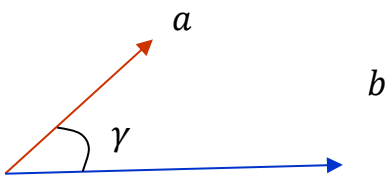
Vector Addition: Adding two vectors

$$c = a + b = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{bmatrix}$$



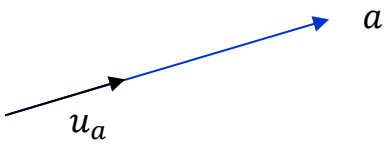
Dot Product: The inner product between vectors

$$a \cdot b = a^\top b = b^\top a = \|a\|\|b\| cos\gamma$$



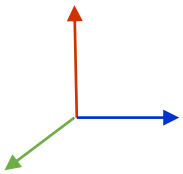
Unit Vector: Vector in the direction of a chosen vector but with magnitude 1

$$u_a = a/\|a\|_2$$

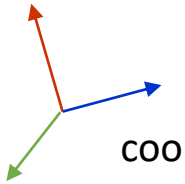


Vectors and Frames of Reference in SE(3)

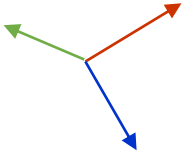
Coordinate Frame: A coordinate system $\{C\}$ comprising a location of the origin $\{x, y, z\}$ and a set of basis vectors $\left(\text{e. g.} \{u_x, u_y, u_z\}\right)$.



World coordinate frame $\{C_o\}$



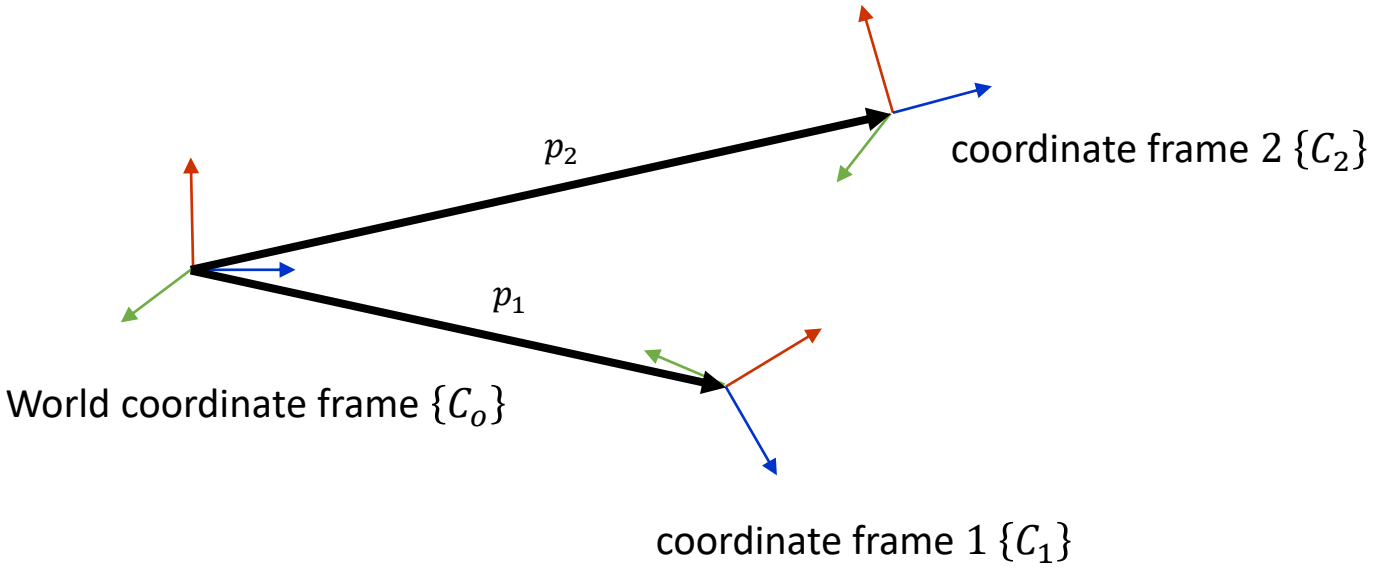
coordinate frame 2 $\{C_2\}$



coordinate frame 1 $\{C_1\}$

A system with several coordinate frames

- The locations of $\{C_1\}, \{C_2\}$ are described by an $\mathbb{R}^3$ position vector, e.g. p_1, p_2
- The orientation of $\{C_1\}, \{C_2\}$ are described by their $\mathbb{R}^3$ basis vectors, e.g. $\{u_{1,x}, u_{1,y}, u_{1,z}\}$ and $\{u_{2,x}, u_{2,y}, u_{2,z}\}$

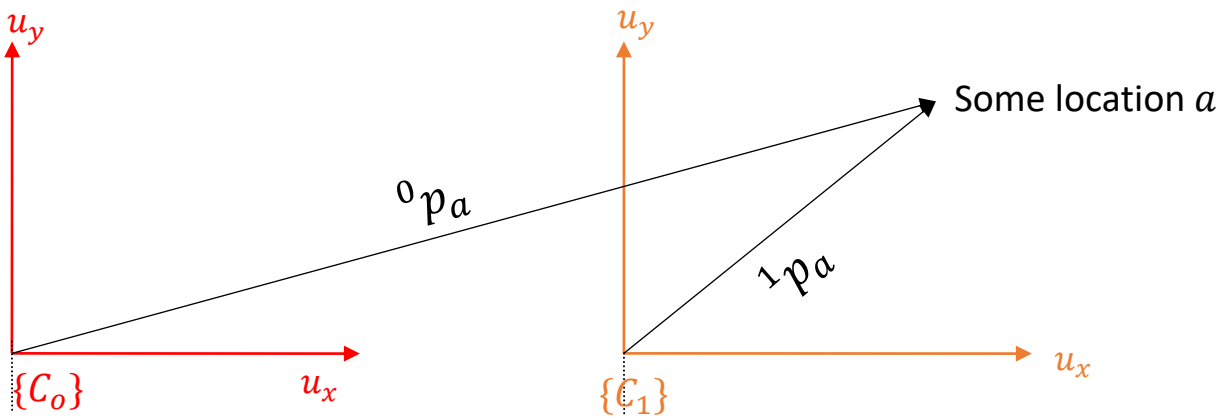


- What are the origin and basis vectors of the world coordinate frame?

Nomenclature for Vectors

Vectors are always described in a specific coordinate frame

- ${}^0p_a$: vector in $\mathbb{R}^3$ pointing to location a in coordinate frame 0
- ${}^1p_a$: vector in $\mathbb{R}^3$ pointing to location a in coordinate frame 1

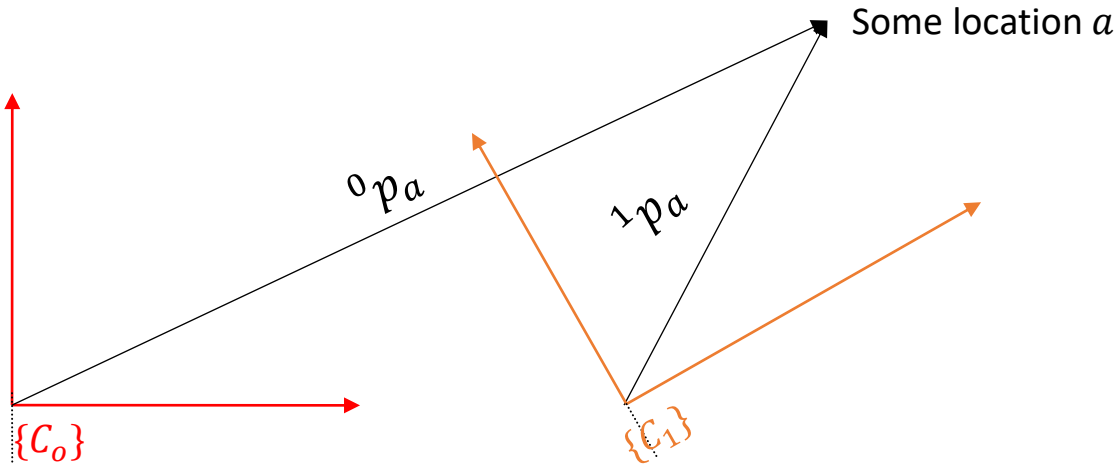


- Vector operations (addition, subtraction, dot product, cross product, etc.) can only applied to vectors in the **SAME** coordinate frame.
- Therefore, **always write down reference frame**!

Moving Vectors Between Coordinate Frames

Given position of a location a in some coordinate frame $\{C_1\}$, ${}^1p_a$,

What is the position of the same point in another coordinate frame $\{C_0\}$, ${}^0p_a$?



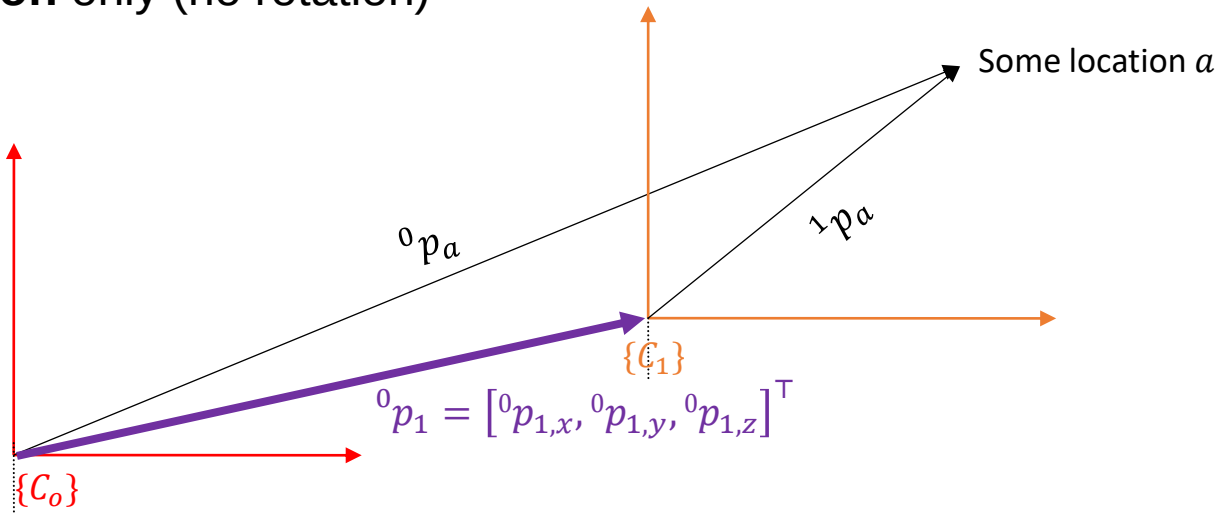
Equation of a position vector described in $\{C_j\}$ relative to $\{C_i\}$:

$$ {}^ip_a = {}^ip_j + {}^i_jR\ {}^jp_a $$

${}^ip_j$: position of origin $\{C_j\}$ in coordinate frame $\{C_i\}$

${}^j_iR$: Orientation of frame $\{C_j\}$ described in coordinate frame $\{C_i\}$

Translation only (no rotation)



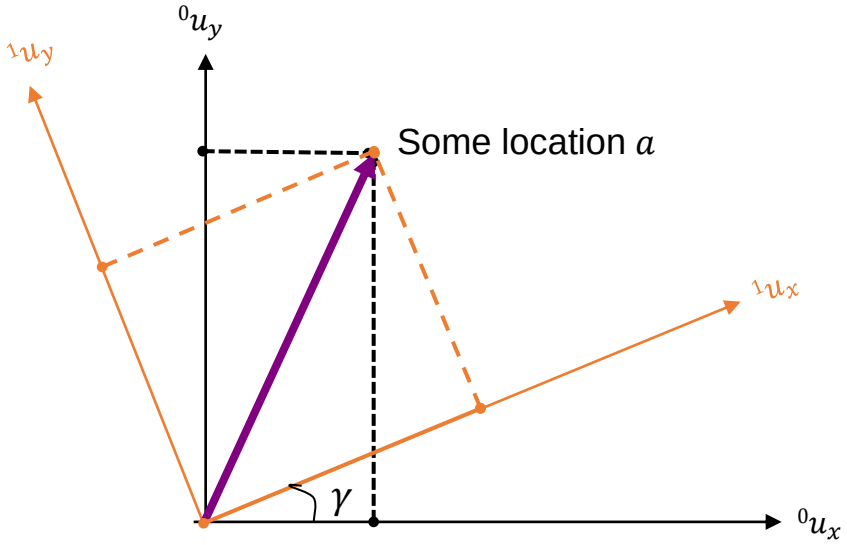
- Apply ${}^i p_a = {}^i p_j + {}^i_j R \, {}^j p_a$ noting that if the orientation of the frames do not change, then ${}^0_1 R = \mathrm{I}$

$$ {}^0 p_a = {}^0 p_1 + {}^0_1 R \, {}^1 p_a = \begin{bmatrix} {}^0p_x \\ {}^0p_y \\ {}^0p_z \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^1p_{a,x} \\ {}^1p_{a,y} \\ {}^1p_{a,z} \end{bmatrix} = {}^0 \begin{bmatrix} p_{1,x} + p_{a,x} \\ p_{1,y} + p_{a,y} \\ p_{1,z} + p_{a,z} \end{bmatrix} $$

- Why is it important to still write out ${}^0_1 R$?

Rotation only (no translation)

- Describe ${}^{1}p_a$ in frame $\{C_0\}$

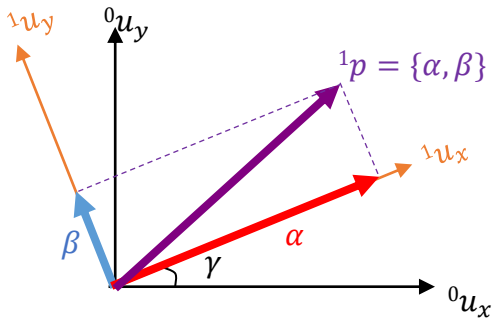


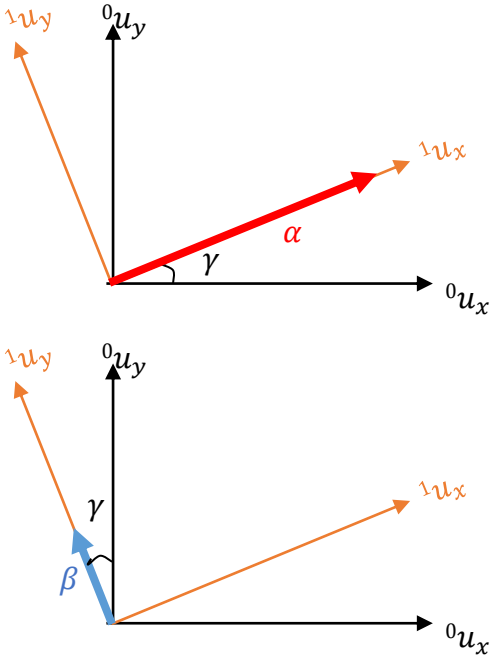
- Apply ${}^{i}p_a = {}^{i}p_j + {}^{i}_{j}R\ {}^{j}p_a$ noting the origins of the frames do not change, i.e. ${}^{0}p_1 = 0$

$$ {}^{0}p_1 = {}^{0}_{1}R\ {}^{1}p_a $$

- Recall ${}^{j}_{i}R$ is orientation of frame $\{C_j\}$ described in coordinate frame $\{C_i\}$
- What is ${}^{0}_{1}R$?

Assume knowledge of ${}^1\boldsymbol{p} = \{\boldsymbol{\alpha}, \boldsymbol{\beta}\}$. Find ${}^0\boldsymbol{p}$.





- The x component of ${}^1p$ in frame $\{C_1\}$ is

$$ {}^0p_x = \begin{bmatrix} \alpha cos(\gamma) \\ \alpha sin(\gamma) \end{bmatrix} $$

- The y component of ${}^1p$ in frame $\{C_0\}$ is

$$ {}^0p_y = \begin{bmatrix} -\beta sin(\gamma) \\ \beta cos(\gamma) \end{bmatrix} $$

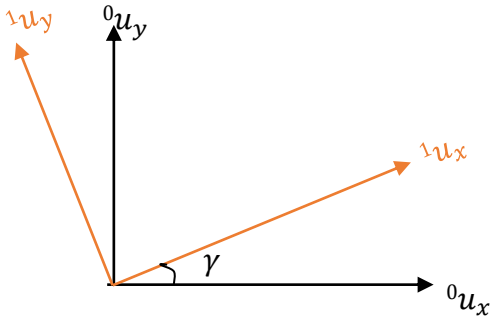
The cumulative vector ${}^0p$ is found by vector addition:

$$ {}^0p = {}^0p_x + {}^0p_y = \begin{bmatrix} -\beta sin(\gamma) \\ \beta cos(\gamma) \end{bmatrix} + \begin{bmatrix} \alpha \cos(\gamma) \\ \alpha \sin(\gamma) \end{bmatrix} = \begin{bmatrix} cos(\gamma) & -\sin(\gamma) \\ \sin(\gamma) & cos(\gamma) \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} $$

Notice how:

$$^0u_{1,x} = \begin{bmatrix} \cos(\gamma) \\ \sin(\gamma) \end{bmatrix}$$

$$^0u_{1,y} = \begin{bmatrix} -\sin(\gamma) \\ \cos(\gamma) \end{bmatrix}$$



Then we can conclude that

$$^0_1R = [^0u_{1,x} \quad ^0u_{1,y}]$$

- Best not to think of R as a rotation *operation*, but as a transformation of perspective.
- All rotation matrices ^i_jR are just the description of bases vectors of frame $\{C_j\}$ in the perspective of frame $\{C_i\}$
- **Useful Tip #1:** Can easily look at rotation matrix and understand how different components are being mapped to a new frame.
- **Useful Tip #2:** Can easily reconstruct rotation matrices if you simply look at each basis vector individually and map them to the new frame.

Useful Rotation Matrices in 2D and 3D about Bases

Rotations in 2D about bases

Z-axis: $R_z(\theta) = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix}$

Rotations in 3D about bases

X-axis: $R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}$

Y-axis: $R_y(\theta) = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$

Z-axis: $R_z(\theta) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

What do these rotations *actually* mean?

Useful Properties:

$$\left({}^0_1R\right)^\top = \left({}^0_1R\right)^{-1} = {}^1_0R$$

$$R^\top R = RR^\top = I$$

$$\det(R) = 1$$

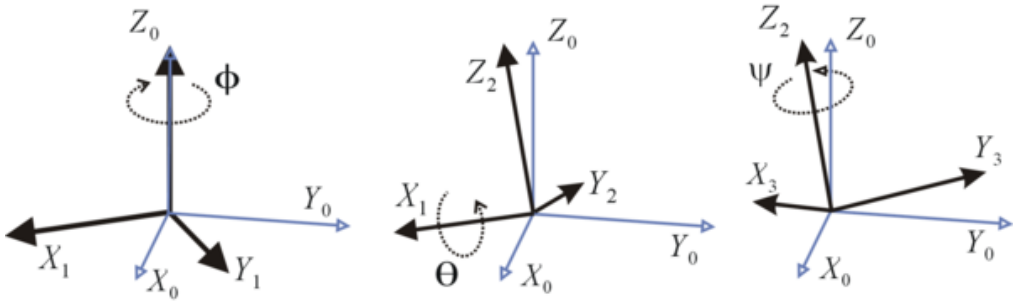
Euler Angles

Any rotation matrix ${}^{i}_{j}R$ may be described by three successive rotations about a set of bases:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix} \qquad R_y(\theta) = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \qquad R_z(\theta) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Intrinsic Rotations: Rotations applied sequentially on the *rotating* frame

e.g. $\phi \rightarrow \theta \rightarrow \psi$:

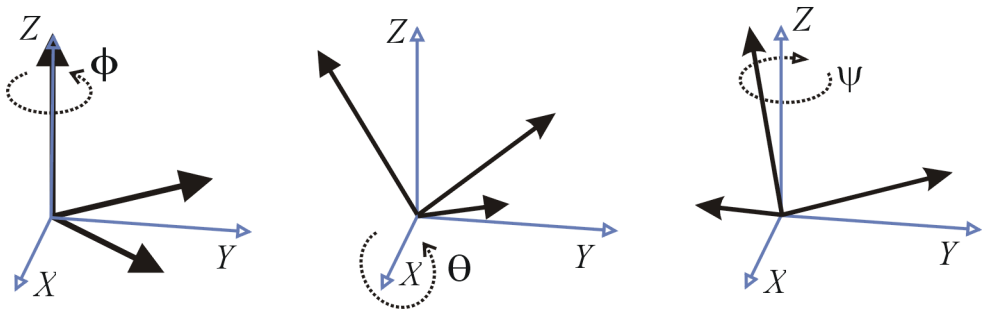


The intrinsic rotation sequence $\phi - \theta - \psi$ describes a rotation:

$$R(\phi, \theta, \psi) = R_z(\psi) R_x(\theta) R_z(\phi)$$

Extrinsic Rotations: Rotations applied on the *static* frame

e.g. $\phi \rightarrow \theta \rightarrow \psi$:



The extrinsic rotation sequence $\phi - \theta - \psi$ describes in a rotation:

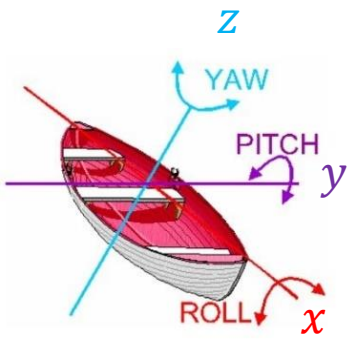
$$R(\phi, \theta, \psi) = R_z(\phi) R_x(\theta) R_z(\psi)$$

Industry nomenclature:

"**Roll-Pitch-Yaw**" – used to describe orientations of aircrafts, boats, computer graphics, etc.

"Rol(r)-Pitch(p)-Yaw(γ)" is equivalent to intrinsic rotations of $x \rightarrow y \rightarrow z$

$$R = R_z R_y R_x$$



- Order is always important, you cannot apply rotations "simultaneously"
- intrinsic or extrinsic is important to know / decide upon.
  - Vehicles often use instrinsics, but robots often use both.

Axis-Angle Representation

Axis angle representation describes a rotation matrix R by a rotation of θ about a SE(3) vector $\boldsymbol{k}$. Avoids confusion of Euler angle order.

- $\boldsymbol{k}$ is a unit vector
- Thus axis angle representation is a 4-element tensor with $\{k_x, k_y, k_z, \theta\}$

$$\boldsymbol{k} = \left[\frac{R_{32} - R_{23}}{2\sin\theta}, \frac{R_{13} - R_{31}}{2\sin\theta}, \frac{R_{21} - R_{12}}{2\sin\theta}\right]$$

$$\theta = \arccos\left(\frac{R_{11} + R_{22} + R_{33} - 1}{2}\right)$$

Rodrigues rotation formula: apply axis angle rotation on vector without constructing R and pre-multiplying.

$$v_{rot} = v\cos\theta + (k \times v)\sin\theta + k(k \cdot v)(1 - cos\theta)$$

- What if you wish to write this as a set of linear algebraic functions? (i.e. no dot product, no cross product?)

Converting SE(3) vector cross products to a matrix multiplication:

$$a = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \qquad a, b \in \mathbb{R}^3$$

$$a \times b = (a \times)b = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} b = \hat{a}b$$

- use the *hat operator* to denote converting SE(3) vector to **skew symmetric matrix.**

Rodrigues formula in Linear Algebra form:

$$v_{rot} = (\cos\theta)v + (\sin\theta)\hat{k}v + (1 - cos\theta)kk^{\top}v$$

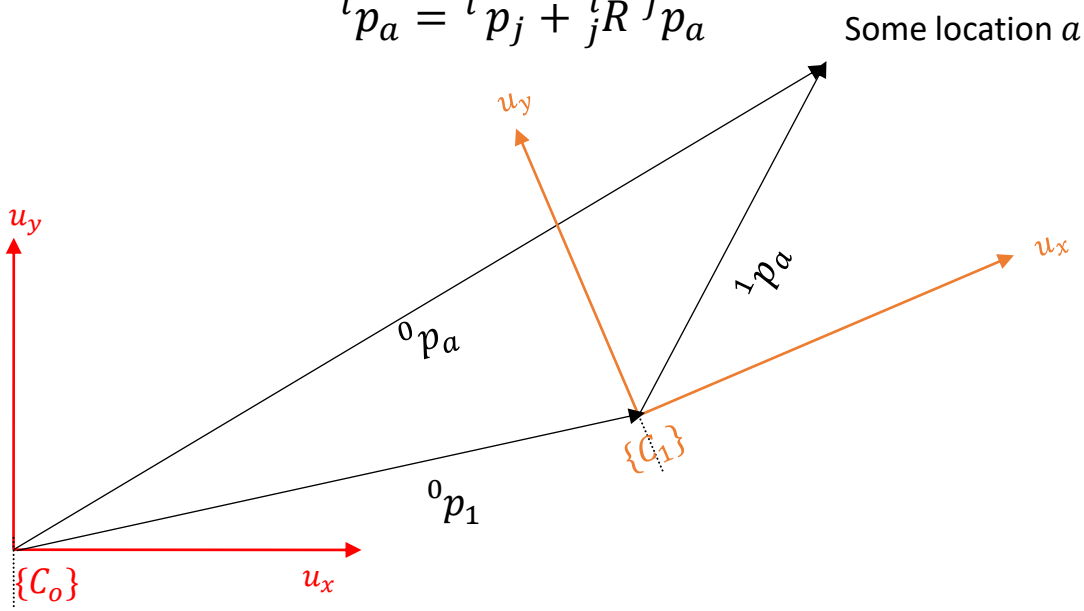
Finally, converting axis angle $\{k, \theta\}$ to rotation matrix R:

$$R = (\cos\theta)I + (1 - \cos\theta)kk^{\top} + \hat{k}\sin\theta$$

Rotation and Translation between Coordinate Frames

With these tools, we have complete described the equations for describing a vector in one coordinate frame when it is known in another coordinate frame

$$^{i}p_a = {}^{i}p_j + {}^{i}_{j}R\ {}^{j}p_a$$



- **Note the order:** Apply rotation matrix to vector first, and then add translation.
  - The nomenclature enforces this: you cannot add vectors in different frames.

$$^{i}p_a = {}^{i}p_j + {}^{i}_{j}R\ {}^{j}p_a$$

Homogenous Matrices in SE(3)

The operation ${}^{i}p_a = {}^{i}p_j + {}^{i}_{j}R\,{}^{j}p_a$ relies on a two-step process.

- Cumbersome to work with
- Equations get messy

The Homogenous Matrix $\boldsymbol{H}$ is a 4×4 matrix is a single **operation** that describes a vector in frame A in a new frame B.

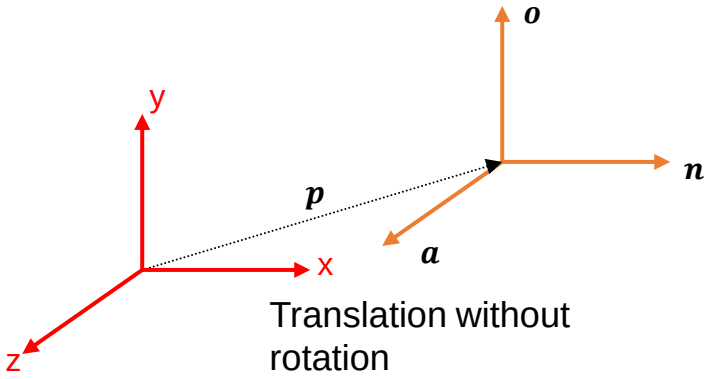
- First, augment position vectors with a trailing 1: $p \Rightarrow \begin{bmatrix} p \\ 1 \end{bmatrix}$
- Then describe the homogenous matrix as

$$ {}^{i}_{j}H = \begin{bmatrix} {}^{i}_{j}R & {}^{i}p_j \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4} $$

The operation for describing vectors between coordinate frames becomes:

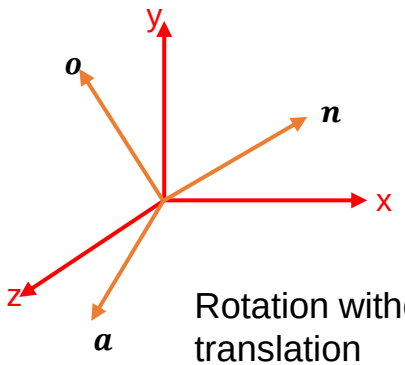
$$ \begin{bmatrix} {}^{i}p_a \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{i}p_j + {}^{i}_{j}R\,{}^{j}p_a \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{i}_{j}R & {}^{i}p_j \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{j}p_a \\ 1 \end{bmatrix} = {}^{i}_{j}H \begin{bmatrix} {}^{j}p_a \\ 1 \end{bmatrix} $$

Examples:



Translation without rotation

$$H_{translation_only} = \begin{bmatrix} 1 & 0 & 0 & p_x \\ 0 & 1 & 0 & p_y \\ 0 & 0 & 1 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

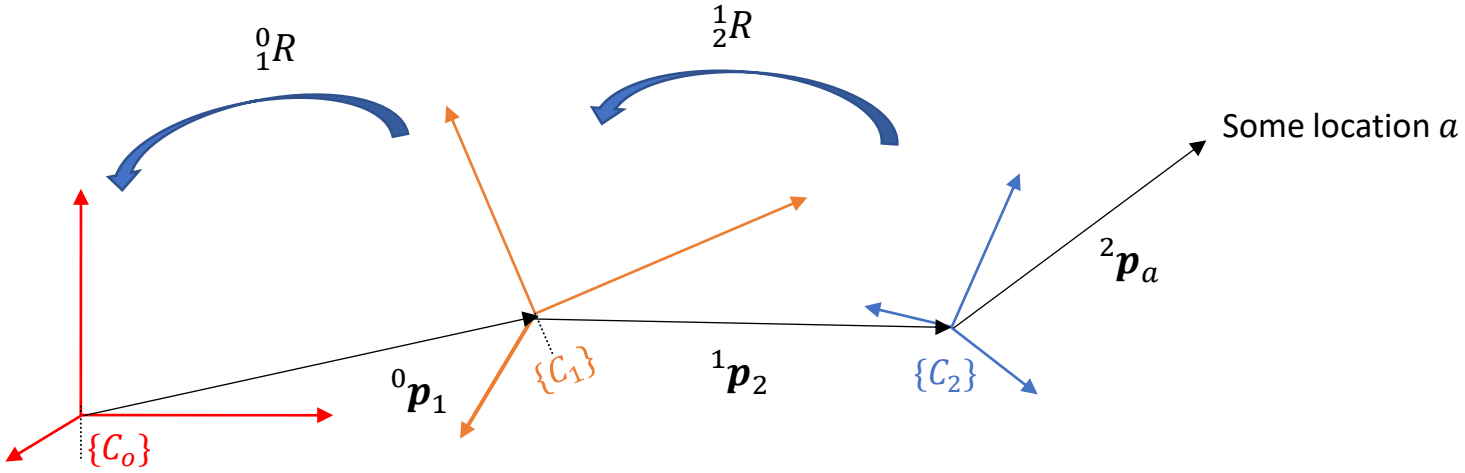


Rotation without translation

$$H_{rotation_only} = \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation could be rotation around z-axis, x-axis, y-axis or any combination

Moving through multiple coordinate frames



- Follow the sequence of homogenous transforms, making note that frames of reference are always correct

$$ {}^0p_a = {}^0p_1 + {}^0_1R\ {}^1p_a = {}^0p_1 + {}^0_1R\left({}^1p_2 + {}^1_2R\ {}^2p_a\right) $$

- With homogenous transforms

$$ \begin{bmatrix} {}^0p_a \\ 1 \end{bmatrix} = {}^0_1H\,{}^1_2H \begin{bmatrix} {}^2p_a \\ 1 \end{bmatrix} = {}^0_2H \begin{bmatrix} {}^2p_a \\ 1 \end{bmatrix} $$